EXPERIMENT 4

CPU SCHEDULING- FCFS, SJF, Priority, Round Robin

AIM:

To implement CPU scheduling algorithms, namely First Come First Serve (FCFS), Shortest Job First (SJF), Priority Scheduling, and Round Robin Scheduling, and calculate the waiting time and turnaround time of processes.

PROCEDURE:

1. Start the Fedora operating system using Oracle VirtualBox.
2. Open the terminal and create a Python program file.
3. Write the programs for the given CPU scheduling algorithms.
4. Execute each program using the Python 3 interpreter.
5. Enter the required process details, including burst time, priority, and time quantum where applicable.
6. Calculate and display the waiting time and turnaround time of each process.
7. Calculate the average waiting time and average turnaround time.
8. Verify the output and record the results.

PROGRAM:

(A) — First Come First Serve (FCFS)

```
bt = []

print("Enter the number of process:")
n = int(input())

print("Enter the burst time of the processes:")
bt = list(map(int, input().split()))

wt = []
tat = []
avgwt = 0
avgtat = 0

wt.insert(0, 0)
tat.insert(0, bt[0])

for i in range(1, len(bt)):
    wt.insert(i, wt[i-1] + bt[i-1])
    tat.insert(i, wt[i] + bt[i])
```

```
    avgwt += wt[i]
    avgtat += tat[i]

avgwt = float(avgwt) / n
avgtat = float(sum(tat)) / n

print("\n")
print("Process\tBurst Time\tWaiting Time\tTurnaround Time")

for i in range(0, n):
    print(str(i) + "\t\t" + str(bt[i]) +
          "\t\t" + str(wt[i]) + "\t\t" + str(tat[i]))

print("\nAverage Waiting time is:", avgwt)
print("Average Turn Around Time is:", avgtat)
```

OUTPUT:

```
[srisha@fedora ~]$ python3 fcfs.py
Enter the number of process:
3
Enter the burst time of the processes:
24 3 3
Process  Burst Time  Waiting Time  Turnaround Time
0     24      0       24
1     3       24      27
2     3       27      30
Average Waiting time is: 17.0
Average Turn Around Time is: 27.0
[srisha@fedora ~]$
```

4(B) – SHORTEST JOB FIRST (SJF)

```
bt = []

print("Enter the number of processes:")
n = int(input())

print("Enter the burst time of the processes:")
bt = list(map(int, input().split()))

wt = [0] * n
tat = [0] * n

# Sort processes based on burst time
processes = sorted(enumerate(bt), key=lambda x: x[1])
```

```
avgwt = 0
avgtat = 0

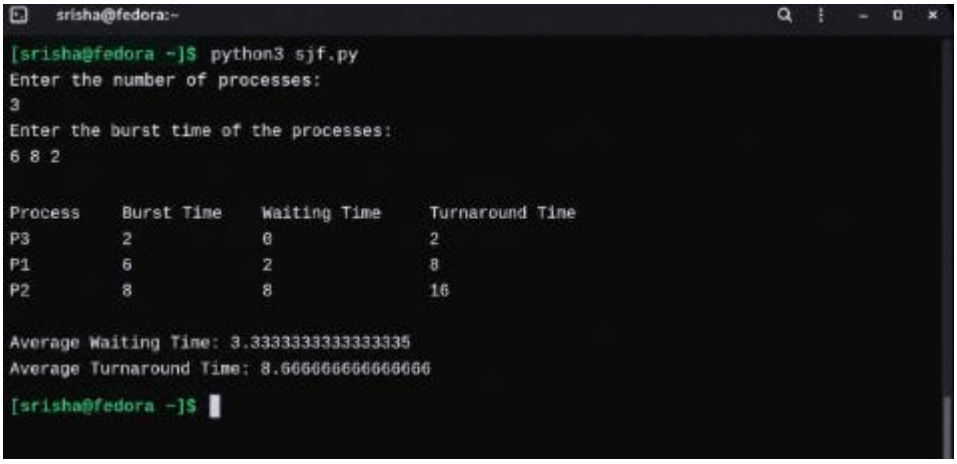
for i in range(n):
    p, burst = processes[i]
    if i > 0:
        wt[i] = wt[i - 1] + processes[i - 1][1]
    tat[i] = wt[i] + burst
    avgwt += wt[i]
    avgtat += tat[i]

print("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time")

for i in range(n):
    p, burst = processes[i]
    print(f"P{p + 1}\t{burst}\t\t{wt[i]}\t\t{tat[i]}")

print("\nAverage Waiting Time:", avgwt / n)
print("Average Turnaround Time:", avgtat / n)
```

OUTOUT:



4(C) – PRIORITY

```
n = int(input("Enter the number of processes: "))

processes = []

for i in range(n):
    bt = int(input(f"Enter burst time for P{i+1}: "))
    priority = int(input(f"Enter priority for P{i+1}: "))
    processes.append([i+1, bt, priority])

# Lower priority number means higher priority
processes.sort(key=lambda x: x[2])

wt = [0] * n
tat = [0] * n
```

```
for i in range(1, n):
    wt[i] = wt[i-1] + processes[i-1][1]

for i in range(n):
    tat[i] = wt[i] + processes[i][1]

print("\nProcess\tBurst\tPriority\tWaiting\tTurnaround")

for i in range(n):

print(f"P{processes[i][0]}\t{processes[i][1]}\t{processes[i][2]}\t\
t{wt[i]}\t{tat[i]}")

print("\nAverage Waiting Time:", sum(wt) / n)
print("Average Turnaround Time:", sum(tat) / n)
```

OUTPUT:

```
[srisha@fedora ~]$ python3 priority.py
Enter the number of processes: 3
Enter burst time for P1: 6
Enter priority for P1: 2
Enter burst time for P2: 8
Enter priority for P2: 3
Enter burst time for P3: 2
Enter priority for P3: 1

Process Burst Priority Waiting Turnaround
P3    2    1     0    2
P1    6    2     2    8
P2    8    3     8    16

Average Waiting Time: 3.3333333333333335
Average Turnaround Time: 8.666666666666666
[srisha@fedora ~]$
```

4(D) – ROUND ROBIN

```
n = int(input("Enter the number of processes: "))

bt = list(map(int, input("Enter burst times: ").split()))
tq = int(input("Enter time quantum: "))

rem_bt = bt.copy()
wt = [0] * n
tat = [0] * n
time = 0

while True:
    done = True

    for i in range(n):
        if rem_bt[i] > 0:
```

```
            done = False

            if rem_bt[i] > tq:
                time += tq
                rem_bt[i] -= tq
            else:
                time += rem_bt[i]
                wt[i] = time - bt[i]
                rem_bt[i] = 0

    if done:
        break

for i in range(n):
    tat[i] = bt[i] + wt[i]

print("\nProcess\tBurst\tWaiting\tTurnaround")

for i in range(n):
    print(f"P{i+1}\t{bt[i]}\t{wt[i]}\t{tat[i]}")

print("\nAverage Waiting Time:", sum(wt) / n)
print("Average Turnaround Time:", sum(tat) / n)
```

OUTPUT:



RESULT:

Thus, the CPU scheduling algorithms FCFS, SJF, Priority Scheduling, and Round Robin were implemented and executed successfully, and the waiting time, turnaround time, and their averages were calculated.